



Multicentric prognostic observational study on biomarker profile in immunotherapy-naïve patients with highly active generalized myasthenia gravis (PROGNO-MG): a study protocol

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Abstract

Background Myasthenia gravis has a variable disease course. Early identification of patients with a highly active disease is crucial to prevent myasthenic exacerbations and crisis, necessitating more aggressive immunotherapy. However, no biomarkers currently exist to identify patients at risk for highly active disease course.

Objective The PROGNO-MG study aims to identify blood-based biomarker signatures associated with a highly active disease course in immunotherapy-naïve patients with acetylcholine-receptor-antibody positive (AChR⁺) generalized Myasthenia gravis (gMG).

Methods This is an investigator-initiated prospective, multicentric, observational study. Seventy newly diagnosed immunotherapy-naïve AChR⁺ gMG patients will be enrolled and followed up for 24 months. Blood samples and clinical data will be collected biannually. Biomarkers (calprotectin, neurofilament light chain, ITIH3, complement activation, kappa free light chains, and exploratory proteome-analyses and immune cell phenotyping) will be measured and their association with clinical outcomes will be evaluated.

Perspective The results will provide evidence for prognostic biomarkers, supporting risk stratification and early individualized treatment for AChR⁺ gMG patients.

Trial registration The study is registered in the German clinical trial register (DRKS00035457).

Keywords Myasthenia gravis · Generalized myasthenia gravis · Prognostic biomarkers · Risk stratification · Acetylcholine receptor antibodies · Proteomics · Immune cell phenotyping · Disease activity · Complement activation

Abbreviations

AChR ⁺	Acetylcholine-receptor-antibody positive	MGC	Myasthenia gravis composite
gMG	Generalized Myasthenia gravis	MGFA	Myasthenia gravis foundation of america
ITIH3	Inter-alpha-trypsin inhibitor heavy chain H3	MG-QoL15r	Myasthenia gravis quality of life 15 revised
LRP4	Lipoprotein receptor-related protein 4	MuSK	Muscle-specific kinase
MG	Myasthenia gravis	PASS	Patient acceptable symptom state
MG-ADL	Myasthenia gravis activities of daily	PRO	Patient-reported outcome
		QMG	Quantitative myasthenia gravis

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Extended author information available on the last page of the article

SSQ Single simple question

Introduction

Myasthenia gravis (MG) is a rare chronic autoimmune disorder marked by muscle weakness associated with fatigability. The clinical course varies widely, ranging from only ocular manifestations to generalized MG (gMG) with recurrent myasthenic crisis characterized by respiratory failure. Despite standard of care, disease burden remains high in many patients, especially with highly active disease course [1]. Consequently, novel therapeutic approaches have emerged, including C5 complement inhibitors and neonatal Fc receptor inhibitors, which have been approved as add-on therapies for patients with highly active acetylcholine-receptor antibody positive (AChR⁺) gMG and in the case of the FcRn inhibitor rozanolixizumab for muscle-specific kinase (MuSK) antibody-positive individuals [2].

The underlying pathophysiology involves autoantibodies targeting antigens at the motor endplate of the neuromuscular junction. IgG-antibodies directed against the acetylcholine-receptor are detected in approximately 80% of all patients, while antibodies against MuSK occur in 3-5% and lipoprotein receptor-related protein 4 (LRP4) account for approximately only 1% of patients. Notably, 10-15% of all patients remain seronegative [1, 3]. Antibody testing is crucial for establishing the diagnosis of MG, as it also influences treatment decisions, such as the consideration of thymectomy and the choice of immunotherapy [3]. However, the role of antibody levels as a biomarker to further guide treatment remain inconclusive and are not useful as a prognostic marker to identify patients at risk for a highly active disease course requiring escalation of immunotherapy to prevent myasthenic crisis or exacerbation [4]. In recent years, several other prognostic biomarkers have been proposed. Serum calprotectin and serum neurofilament light chain levels were shown to be significantly increased in MG patients, however, mild correlation to disease activity in a large patient population was only seen for calprotectin [5, 6]. Recently, proteomics-based approaches were useful in identifying single potential biomarker, e.g., inter-alpha-trypsin inhibitor heavy chain H3 (ITIH3) in AChR⁺ patients that correlated well with clinical disease activity [7], as well as clustering for proteome signatures that predict treatment response in subsets of MG patients [8]. In addition, comparative proteomic analyses have been useful in identifying additional candidate biomarkers to distinguish AChR⁺ MG from other neuroinflammatory diseases, further supporting the utility of proteomics-based approaches for biomarker discovery in MG [9]. Moreover, immune cell phenotyping and complement activation profiles have shown promise in

exploratory studies [10–12]. Importantly, most existing biomarker studies in MG have been cross-sectional or included treatment-exposed populations, limiting their prognostic applicability. In this study, we seek to evaluate the prognostic value of these proposed biomarkers in a subset of AChR⁺ gMG patients with a highly active disease course, potentially supporting clinical decision-making and personalized care.

Methods

Aim of the trial

PROGNO-MG hypothesizes that gMG patients who will progress to a highly active disease variant show a distinct immune and biomarker signature prior to progression compared to patients showing a moderate disease course. This might enable personalized treatment decision in the future.

Study description and study design

PROGNO-MG is a prospective investigator-initiated, multicentric, non-interventional observational cohort study. The study is conducted across specialized MG centers in Germany. All enrolled patients will be followed for 24 months with standardized assessments every 6 months (Fig 1). PROGNO-MG is an observational study and participation will not impact on treatment choices or concomitant care.

Eligibility criteria

Inclusion criteria are as following:

- Written informed consent
- Age 18-75 years
- Newly diagnosed AChR⁺ gMG (diagnosis <12 months) with one of the following: response to an acetylcholinesterase inhibitor or decrement after repetitive serial stimulation (outside M. orbicularis oculi) or increased jitter in single-fiber EMG
- Immunotherapy-naïve (corticosteroids and symptomatic therapy with pyridostigmine allowed)

Exclusion criteria:

- Prior immunosuppressive therapy (including azathioprine, mycophenolate mofetil, methotrexate, cyclosporine, tacrolimus) or targeted escalation therapies (including C5-inhibitor, FcRn-inhibitor, B-cell depletion like rituximab)
- Prior thymectomy or thymoma-associated MG

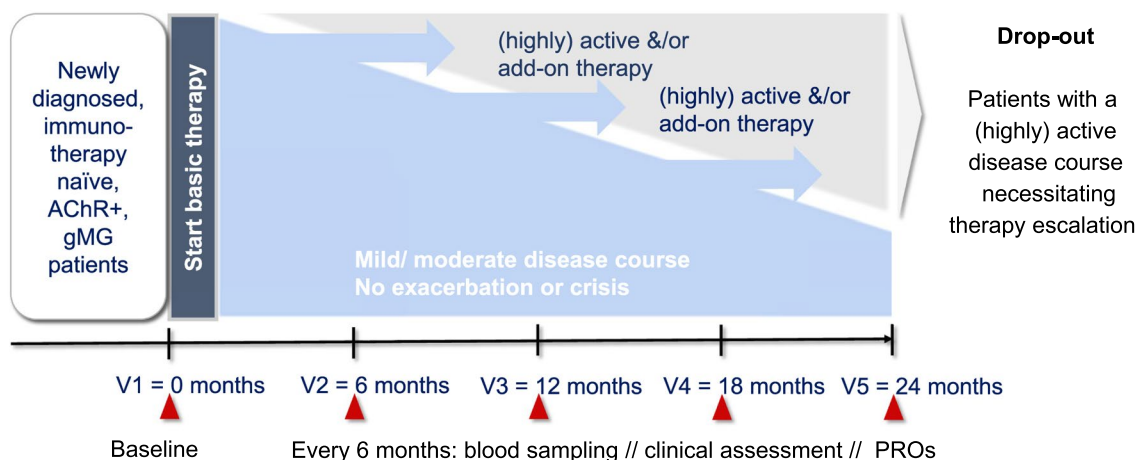


Fig 1. PROGNO-MG Study flow chart. Biannual study visits are planned for acetylcholine-receptor antibody positive (AChR⁺) generalized Myasthenia gravis (gMG) patients. Progression to a highly

active disease course will lead to a drop-out and an unscheduled visit with clinical assessment, blood sampling and patient-reported outcome (PRO) assessment

- Recent (≤ 3 months) use of intravenous immunoglobulins, plasma exchange, or immunoadsorption

Arms and interventions

This prospective observational study will enroll 70 patients with gMG. Based on disease activity trajectories, we anticipate 10–15% of the participants will develop highly active disease within an observation period of two years. We will compare these individuals to the remaining 85–90% exhibiting a moderate disease course to identify associated biomarker signatures. Immunotherapy-naïve was defined as absence of prior treatment with long-term immunosuppressive or targeted immunotherapies. Corticosteroid treatment was permitted, as it represents standard first-line therapy in newly diagnosed generalized MG, and withholding treatment for study inclusion would not be ethically appropriate.

According to the German guideline for diagnosis and treatment of myasthenic syndromes, a highly active gMG, including so-called therapy-refractory MG were defined as [3]:

1. A moderate to high MGFA classification ($\geq$ MGFA IIb) and/or at least two recurrent severe exacerbations or myasthenic crises requiring therapeutic intervention (e.g. IVIG, plasma exchange, or immunoadsorption) within one year of diagnosis, despite adequate disease-modifying and symptomatic treatment; **or**
2. Persistent, functionally relevant symptoms ($\geq$ MGFA IIa) along with at least one severe exacerbation or myasthenic crisis within the past year, despite adequate disease-modifying and symptomatic treatment; **or**

3. Persistent, functionally relevant mild to moderate symptoms ($\geq$ MGFA IIa) for more than two years, despite adequate disease-modifying and symptomatic treatment.

Statistical power was calculated to detect hazard ratios ≥ 3 for biomarkers predicting disease progression.

Patients will undergo a standardized assessment at baseline and every 6 months for overall 24 months (Table 1):

- Clinical assessments (e.g. MGFA [13], QMG [14], MGC scores [15]), including current immunotherapy and symptomatic therapy with reported side effects, thymectomy, comedications and comorbidities
- Patient-reported outcomes (MG-ADL [16], MG-QoL15r [17], Neuro-QoL Fatigue [18], Beck Depression Inventory [19], PASS [20], SSQ [21])
- Sampling for biomarker analysis
- Biomarker testing: calprotectin, neurofilament light chain, ITIH3, complement activation profiles, kappa free light chains
- Exploratory analyses: proteomics and immune cell phenotyping

At baseline additional demographic data will be collected, including sex, age, body mass index, vaccination status and potential substance abuse. Unscheduled visits will be conducted in the event of disease exacerbations.

Outcome measures

Primary endpoint is the development of a highly active disease course requiring escalation to add-on immunotherapy. Potential prognostic biomarkers (calprotectin, neurofilament light chain, ITIH3, kappa free light chains) as well

Table 1 Summary of standardized assessment

Assessment	V1 (baseline)	V2 (6 months)	V3 (12 months)	V4 (18 months)	V5 (24 months)	Drop-out before regu- lar study end
Clinical parameters (height, weight, ethnicity, family history, substance history, vaccination status)	X					
Clinical scores (QMG, MGFA, MGC)	X	X	X	X	X	X
Thymectomy status	X	X	X	X	X	X
Survey on adverse events		X	X	X	X	X
MG-specific therapy	X	X	X	X	X	X + planned intensified therapy
Comorbidities	X	X	X	X	X	X
Concomitant therapies	X	X	X	X	X	X
Patient reported outcomes (MG-ADL, MG-QoL15; Neuro-QoL fatigue score, BDI, PASS, SSQ)	X	X	X	X	X	X
Biosampling (blood sampling)	X	X	X	X	X	X

MG = myasthenia gravis; QMG = quantitative myasthenia gravis score; MGFA = myasthenia gravis foundation of America clinical classification; MGC = myasthenia gravis composite score; MG-ADL = myasthenia gravis activities of daily living; myasthenia gravis quality of life revised; Neuro-QoL fatigue score = quality of life in neurological disorders; BDI = Beck's depression inventory; PASS = patient acceptable symptom state; SSQ = single simple question

as temporal differences for immune cell phenotyping, proteome analyses and complement activation profiles (circulating complement activation products and regulatory proteins, including C3b, iC3b, C4b, sC5b-9, Factor H, and Factor I) will be characterized. Secondary endpoints are demographic and clinical differences between highly active and moderate disease course and to assess the risk for myasthenic exacerbation/crisis of gMG patients within two years.

Data and biosample management

Patient data will be pseudonymized locally on each study site and then be handled in electronic case report forms and submitted electronically via the open-source platform RED-Cap. The data monitoring committee is represented by F.S. and A.M., and will work independent of the sponsor. The sponsor of the study has no access to individual patient data. Pseudonymized biosamples will be forwarded for further processing to a biobank, enabling standardized processing and storage until further analysis. Standard operation protocols for biosample handling and shipping are implemented and followed by each study site. Missing data is handled based on the performed analysis, e.g., imputation for proteome analysis.

Statistical analysis

The study is exploratory, and the number of cases is not sufficient to answer confirmatory questions. The study only serves to discover indications of possible correlations between biomarker levels and outcomes. Overall patient sample size is estimated based on a 10-15% of gMG patients progressing to a highly active disease course within

two years for this explorative study [2, 3]. The sample size was calculated using a Cox regression for a *log Hazard Ratio on a covariate* with a standard deviation of 1. A sample size of $n=70$ provides a power of 81% for a regression coefficient of ≥ 1.099 (corresponding to a hazard ratio of ≥ 3) at a two-sided significance level of 5%. The calculation assumes an event rate of 12% ($n = 8-9$). The estimation accounts for potential confounding covariates in a multiple regression analysis, assuming an R^2 of 0.2. The software nQuery Advisor 9 was used for sample size estimation. Cox proportional hazards models will be used to assess the relationship between biomarkers and disease progression. Time-dependent covariates and potential confounders will be adjusted in Cox regression models. Proteomic data will be analyzed using false discovery rate adjustments for multiple testing. Statistical analysis will be performed by the Institute of Biometry and Clinical Epidemiology Charité - Universitätsmedizin Berlin.

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Perspective

PROGNO-MG will provide important, prospective clinical data on AChR⁺ gMG with highly active disease course and allow the evaluation of proposed biomarkers (calprotectin, neurofilament light chain, ITIH3, complement activation, kappa free light chains) in a well-defined immunotherapy-naïve gMG subset. It will further evaluate temporal immune

cell phenotypes and proteomic profiles of these patients. Early identification of patients at risk for a highly active disease course may enable targeted therapeutic approaches to improve patient outcomes and reduce disease burden.

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Author contributions F.S., A.M., and H.W. conceptualized and designed the study. F.K., F.F.K. and F.S. wrote the manuscript. All authors revised the manuscript for intellectual content.

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Data availability Anonymized study data will be available upon reasonable request following publication of the study results.

Declarations

Ethics approval The study is conducted according to the Declaration of Helsinki and approved by the Ethics Committee of the Charité – Universitätsmedizin Berlin (EA2/195/24); written informed consent will be given by all participants. The study is registered in the German clinical trial register (DRKS00035457, retrospectively registered on the 20.01.2025).

Protocol amendments Version 1.0 of the study protocol. Any changes to the study protocol will be communicated immediately to all investigators and participants. Next to an update of the study registry, the investigators will publish an updated version of this manuscript.

Consent for publication Study results will be published in aggregated form without identifying information.

Competing interests F.K. received honoraria for lectures from argenx and travel compensation from Alexion. F.F.K. received honoraria for lectures and travel compensation from Alexion, argenx, Merck, Novartis, and Takeda. He received research support as fellow of the German Research Foundation (DFG)- and Hannover Medical School (MHH)-funded Clinician Scientist Program (PRACTIS) at MHH and from Merck, Siemens, and Erwin-Röver-Foundation. S.G. received travel/accommodation/meeting expenses from Alexion Pharmaceuticals, argenx, Bayer, Grifols, Octapharma, and UCB pharma and received speaker honoraria from Alexion Pharmaceuticals, argenx, and UCB pharma. P.T. received travel/accommodation/meeting expenses and honoraria for a lecture by Astra Zeneca Rare Disease. C.S. received speaker honoraria from Alexion, argenx, Desitin, and TAD and received personal fees for advisory boards and grants for congress travel and participation from Alexion and argenx. J.Z. received travel/accommodation/meeting expenses from Alexion Astra Zeneca Rare Disease, argenx, Johnson&Johnson, Kedrion, Sanofi and UCB. She also received speaking honoraria and honoraria for attendance at advisory boards from Alexion Astra Zeneca Rare Disease, Amicus, argenx, Kedrion, Roche, Sanofi, StreamedUp, and UCB. She received financial research support (paid to her institution) from argenx and UCB. J.Z. is board member of the German Myasthenia Association and member of the German Muscle Society. J.P.S. received lecture and consulting fees from Alexion, argenx, Axunio, Hormosan, Jans-

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